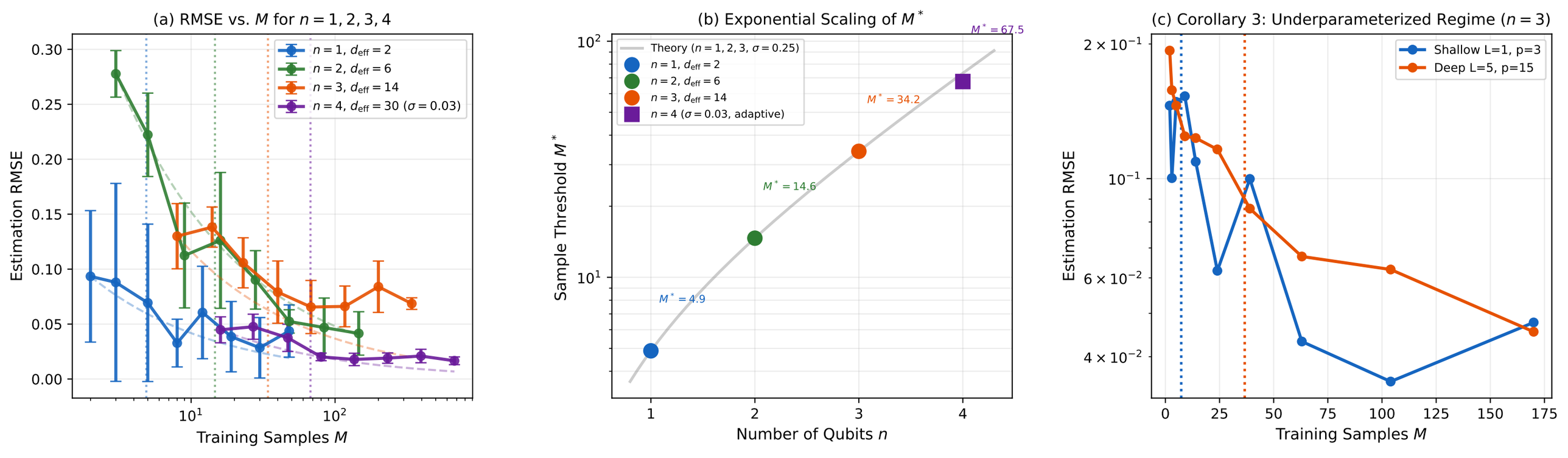
(a) RMSE vs. M for $n = 1, 2, 3, 4$
Estimation RMSE
Training Samples M
$n = 1$, $d_{\text{eff}} = 2$
$n = 2$, $d_{\text{eff}} = 6$
$n = 3$, $d_{\text{eff}} = 14$
$n = 4$, $d_{\text{eff}} = 30$ ($\sigma = 0.03$)
(b) Exponential Scaling of M^*
Sample Threshold M^*
Number of Qubits n
Theory ($n = 1, 2, 3$, $\sigma = 0.25$)
$n = 4$ ($\sigma = 0.03$, adaptive)
$M^* = 4.9$
$M^* = 14.6$
$M^* = 34.2$
$M^* = 67.5$
(c) Corollary 3: Underparameterized Regime ($n = 3$)
Shallow L=1, p=3
Deep L=5, p=15